final Exam

Second Year

Classical Mechanics I

University Of Tobruk

Facility Of science

Medical physics

Q1. Choose the correct answer from the following:

1. In an elastic collision, which quantities are conserved?

A) Momentum only

B) Energy only

C) Momentum and kinetic energy

D) Potential energy only

2. An inelastic collision is defined as one in which:

A) Only kinetic energy is conserved

B) Momentum is not conserved

C) Kinetic energy is not conserved but momentum is

D) No quantity is conserved

3. The formula for the center of mass of a system is:

A) $R = \Sigma r / \Sigma m$

B) $R = \Sigma(mr) / \Sigma m$

C) $R = \Sigma m / r$

D) $R = r - m$

4. The motion of a rigid body can be separated into:

A) Rotational only

B) Translational only

C) Translational & rotational

D) Vertical & horizontal

5. Rotational kinetic energy of a rigid body is given by:

A) $K = 1/2\ mv^2$

B) $K = 1/2\ I\omega^2$

C) $K = I\omega$

D) $K = m\omega^2$

6. What is the SI unit of linear momentum?

A) N·s

B) kg·m/s

C) m/s^2

D) J·s

7. Newton's second law for rotational motion is:

A) $\tau = I\omega$

B) $\tau = I\alpha$

C) $F = ma$

D) $\tau = mr^2$

8. Torque is the rotational equivalent of:

A) Velocity

B) force

C) acceleration

D) energy

9. Total kinetic energy of a rolling body is:

A) Sum of translational and rotational B) Only translational

C) Zero D) Only rotational

10.For a solid disk, the moment of inertia about its center is:

A) MR^2 B)1/2 MR^2

C)2/5 MR^2 D)1/12 MR^2

Q2: A) define: a) angular velocity. b) angular acceleration.

B) define linear momentum and give its unit.

Q3. Discuss the collision and their types.

Q4. A solid sphere of mass 2 kg and radius 0.2 m rolls without slipping with speed 4 m/s. Find its total kinetic energy.

Q5:(A) A 5 kg firecracker explodes into two parts: 2 kg moves at 4 m/s east. Find the velocity of the other part.

(B). Three particles of masses 1 kg, 2 kg, 3 kg at x=0,2,4 m. Find center of mass.

Q6:(A) Initial velocity of object 1: u1 = (3, 4, 0) Final velocity of object 1. 2kg

(B). A force of 10 N acts perpendicular to a rod at a distance of 0.5 m from the axis. Find the torque.

Good luck

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